

# Space-charge distortion of ionization profile monitor measurements in the CSNS rapid-cycling synchrotron

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## Abstract

The ionization profile monitor (IPM) of the CSNS rapid-cycling synchrotron records the transverse proton distribution by collecting residual-gas electrons or ions. During their drift to the collector the secondaries are deflected by the bunch space-charge field, so that the detected r.m.s. width differs from the beam size. Particle tracking with Virtual-IPM 2.3.1 is used to quantify that error for the CSNS geometry and to assess the available mitigations. In electron mode a guiding field of 300 G reduces the relative expansion to the one-percent level for all beam families, intensities, cage voltages and orbit offsets examined; the design field of 0.1 T is conservative. In ion mode the same-sign kick always inflates the profile: at 100 kW a 10 mm beam is expanded by 92% ( $\text{H}_2^+$ , injection) and 104% ( $\text{H}_2^+$ , extraction). Raising the cage voltage from 5 to 30 kV reduces but does not remove the bias. The true size can be recovered by inverting a species-resolved look-up of the expansion against bunch charge and cage voltage. The recommended operating point is electron collection at  $B \gtrsim 300$  G.

## Keywords

ionization profile monitor; space charge; CSNS RCS; electron collection; beam-size measurement

## 1 Introduction

An ionization profile monitor infers the transverse beam density from residual-gas ionization products collected on a position-sensitive detector [1, 4]. The CSNS rapid-cycling synchrotron (RCS) IPM [2, 3, 4, 5] uses a  $220 \times 231$  mm cage biased at a design voltage of 25 kV ( $E_y \simeq 108 \text{ kV m}^{-1}$ ) and a magnet specified at 0.1 T (available to 0.2 T). Electrons are collected at the top electrode; the three ion species resolved in the time-of-flight spectrum— $\text{H}_2^+$ ,  $\text{H}_2\text{O}^+$  and  $\text{N}_2^+$ —are collected at the bottom. Two proton bunches circulate. At 100 kW each carries  $N = 7.8 \times 10^{12}$  particles (scaled linearly with beam power  $P$ ). Injection is at 80 MeV ( $\beta = 0.39$ ,  $\sigma_t = 120$  ns, painted  $\sigma_0 \simeq 25$  mm or a 10 mm core); extraction is at 1.6 GeV ( $\beta = 0.93$ ,  $\sigma_t = 20$  ns,  $\sigma_0 \simeq 10$  mm).

While the secondaries drift they experience the bunch electric field and the Lorentz-boosted bunch magnetic field. The detected r.m.s. width  $\sigma_m$  is therefore not the beam width  $\sigma_0$ . The same mechanism has been analysed for magnetic electron IPMs by Vilsmeier, Sapinski and Storey [7] and for ion IPMs by Shiltsev [8] and by the ISIS group [9]. The question for CSNS is quantitative: how large is the error, how does it grow with bunch charge and shrink with beam size, and how efficiently do the available mitigations—guiding  $B$ , cage voltage, and a species-resolved correction—remove it?

The calculation uses Virtual-IPM 2.3.1 [6] with uniform cage fields and  $10^5$  secondaries per case. Distortion is isolated by comparing bunch space charge on and off at fixed guiding  $E$  and  $B$ ,

$$\Delta = \frac{\sigma_{\text{SC}} - \sigma_{\text{off}}}{\sigma_{\text{off}}}. \quad (1)$$

A positive  $\Delta$  is an inflated size; a negative  $\Delta$  is compression. Microchannel-plate saturation, electromagnetic interference, secondary electrons from the ion trap and the measured cage map are not included. Those hardware effects, rather than space charge, are identified in Refs. [2, 3, 5] as the present electron-mode limit once a sufficient guiding field is applied.

## 2 Space-charge interaction with collected secondaries

Characteristic times fix the regime. At 25 kV an electron crosses the 115.5 mm half-gap in  $\tau \simeq 3.5$  ns. The same path takes  $\tau_2 \simeq 210, 631$  and  $787$  ns for  $\text{H}_2^+$ ,  $\text{H}_2\text{O}^+$  and  $\text{N}_2^+$ . The time to leave a 10 mm beam is  $\tau_0 \simeq 74, 221$  and  $275$  ns. Injection bunches (120 ns) are longer than  $\tau_0$  for  $\text{H}_2^+$ ; extraction bunches (20 ns) are shorter than every  $\tau_0$ . The bunch spacing is 980 ns at injection and 409 ns at extraction.

Electrons are attracted to the proton bunch. A moderate kick focuses them towards the axis (compression). When the bunch field exceeds the cage field they cross the axis and the collected profile is wider than the beam (over-focusing). Peak transverse bunch fields on a 10 mm beam are approximately  $29 \text{ kV m}^{-1}$  (injection) and  $73 \text{ kV m}^{-1}$  (extraction) at 100 kW, versus 145 and  $363 \text{ kV m}^{-1}$  at 500 kW; the cage field is  $108 \text{ kV m}^{-1}$ . At 500 kW extraction electrons are trapped in the beam potential during the passage, the regime identified in Ref. [7] as requiring a magnetic field rather than an increase of  $V$ .

A guiding field converts the kick into cyclotron motion. If the impulse is delivered early in the flight, the displacement at the collector is proportional to  $\sin(\omega_c \tau)/\omega_c$  and vanishes whenever  $\omega_c \tau = n\pi$ . Successive zeros of  $\Delta(B)$  are therefore spaced by

$$\Delta B = \frac{\pi m_e}{e\tau} \propto \sqrt{V}. \quad (2)$$

The envelope of the oscillation falls as  $B$  increases. Magnetic mitigation is the decay of that envelope, not the choice of a particular zero. The same cyclotron-phase picture is the “one gyration” tuning rule used at GSI and J-PARC [10, 11].

Ions have the same sign as the beam, so the kick is always outward and  $\Delta$  is always positive. In the impulsive limit (bunch length  $\ll \tau_0$ ) the transverse impulse scales as  $N/\sigma_0^2$  and the subsequent drift as  $\tau_2 \propto 1/\sqrt{V}$ , giving [8]

$$h - 1 \equiv \Delta \propto \frac{N}{V \sigma_0^2}, \quad (3)$$

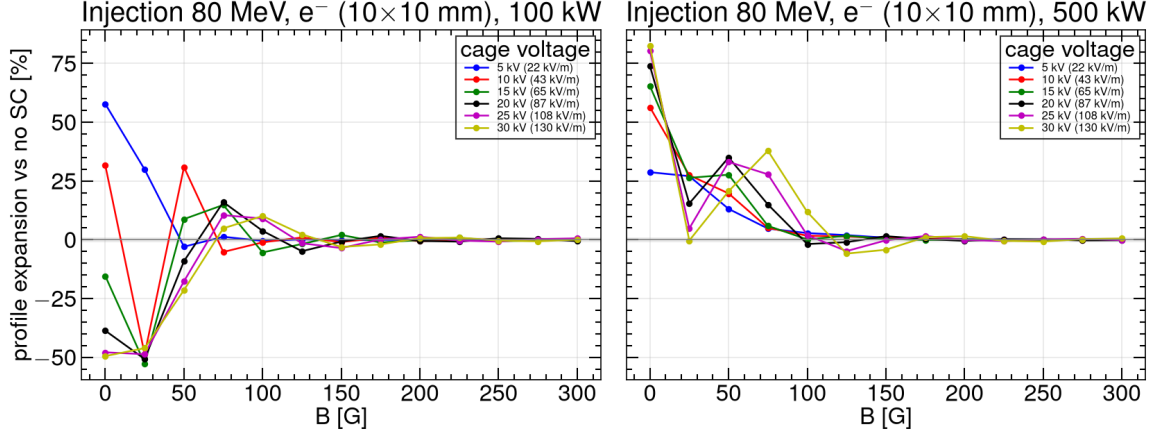
with a mass dependence  $\propto M^{-1/2}$ . CSNS occupies a mixed regime: extraction is impulsive, whereas at injection  $\text{H}_2^+$  leaves the beam while the 120 ns bunch is still passing. Magnetic confinement is an electron-mode instrument [7, 8] and is not treated further as an ion-mode correction.

## 3 Profile distortion and growth of the measured size

### 3.1 Electron collection without a guiding field

At  $B = 0$  the collected electron width is not a reliable size. On the 10 mm injection beam at 100 kW,  $\Delta$  changes from expansion to compression at 13 kV (+9% at 12 kV,  $\approx 0$  at 13 kV,  $-8\%$  at 14 kV). At 500 kW the same voltage scan never changes sign:  $\Delta$  remains positive and grows with  $V$  from +29% at 5 kV to +83% at 30 kV (Fig. 1). That is the trapping regime of Section 2: raising  $V$  does not un-trap a 500 kW bunch. The same intensity-dependent sign reversal is reported at GSI (electrons shrink without  $B$  at moderate intensity) [10] and at J-PARC (a factor-of-two shrink in electron mode) [11].

The map  $\sigma_m(\sigma_0)$  is not monotonic at  $B = 0$ : a small beam may appear larger or smaller than a large one, according to whether the kick focuses or over-focuses. The Voitkiv  $\text{H}_2$  double differential cross-section used by Virtual-IPM contributes a  $\sigma$ -independent quadrature smear of 2.9–3.2 mm ( $\approx 2$ –2.5 eV per transverse axis), which is +40–48% at  $\sigma_0 = 3$  mm and only +1% at 20 mm. One cyclotron period removes that smear (Section 4.1).



**Fig. 1:** Relative expansion  $\Delta$  of the electron profile as a function of guiding field for cage voltages from 5 to 30 kV, on the 10 mm injection beam. At  $B = 0$  the 100 kW family crosses zero; the 500 kW family does not.

Along the energy ramp at  $B = 0$ , a long bunch (120 ns) is compressed and a short bunch (20 ns) is inflated. Both errors become milder as  $\beta$  rises, because the electrons spend less time in the bunch: at 100 kW the 120 ns family goes from  $-46\%$  (200 MeV) to  $-32\%$  (1.6 GeV), and the 20 ns family from  $+77\%$  (80 MeV) to  $+38\%$  (1.2 GeV). Energy alone does not restore the profile.

### 3.2 Ion collection

Every ion case has  $\Delta > 0$ . Figure 2 shows  $\sigma_m$  versus  $\sigma_0$  at the design 0.1 T and 100 kW: every species and both rings lie above the diagonal. Figure 3 shows the growth with bunch charge and the decrease with beam size required by Eq. (3).

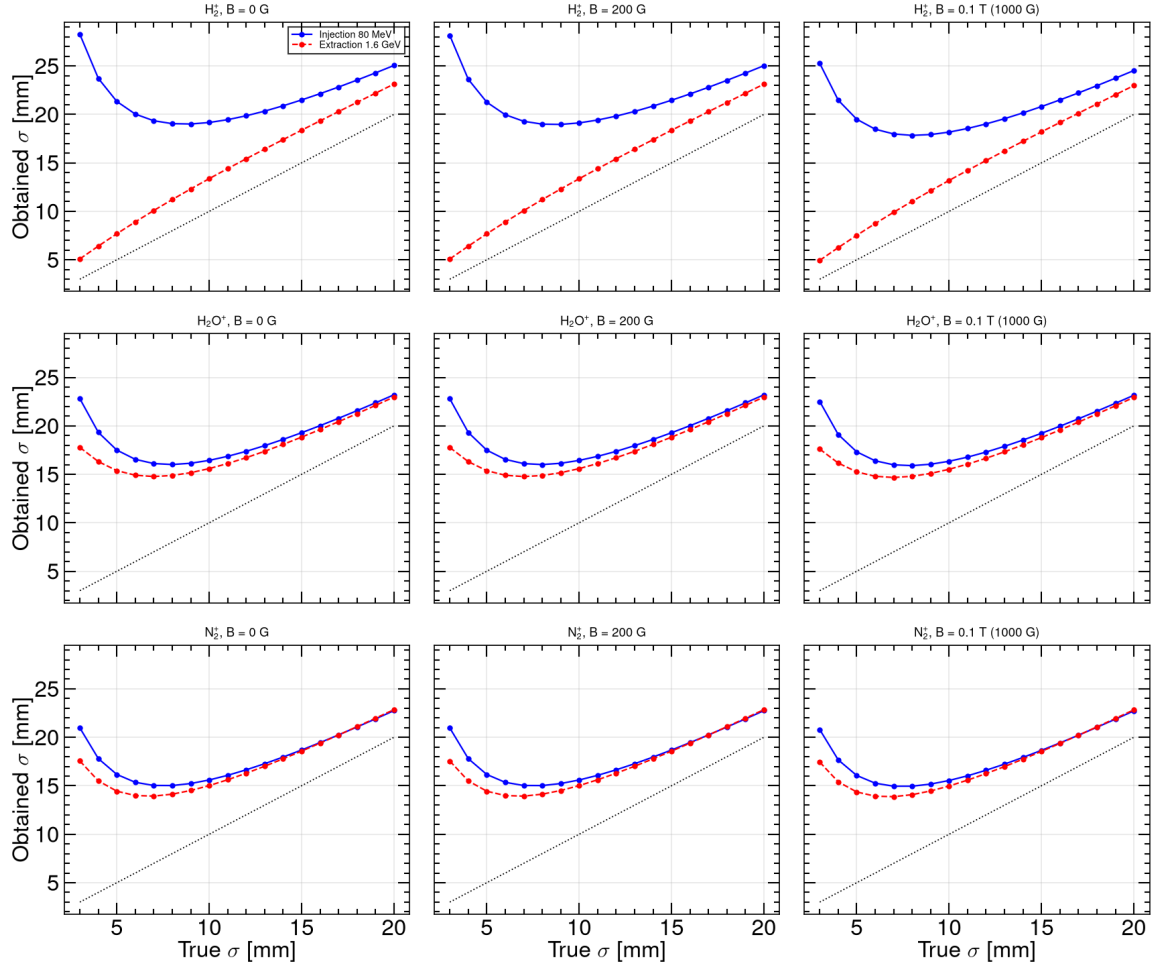
For  $H_2^+$  at injection,  $\sigma_0 = 10$  mm and 0.1 T, the expansion grows from  $+82\%$  (18.2 mm) at 100 kW to  $+175\%$ ,  $+279\%$ ,  $+390\%$  and  $+422\%$  (52.1 mm) at 200, 300, 400 and 500 kW. Between 100 and 400 kW the growth is  $\propto P^{1.12}$ , consistent with the linear kick plus the second-order term of  $\sigma_m^2 = \sigma_0^2 + \kappa N + \dots$  [8]. At 500 kW the profile reaches the 110 mm half-cage and the exponent saturates.

At extraction, with ions born in time with the bunch, 10 mm, 100 kW and 25 kV, the expansions are  $+104\%$  ( $H_2^+$ ),  $+41\%$  ( $H_2O^+$ ) and  $+40\%$  ( $N_2^+$ ) at  $B = 0$ , in agreement with a reduced line-charge integration of the same equations to  $\pm 1\%$  absolute (Fig. 4; model values  $+102\%$ ,  $+40\%$  and  $+39\%$ ). The 20 ns bunch is impulsive, so  $H_2^+$  is the most distorted species. At 500 kW the same family reaches  $+408\%$ ,  $+243\%$  and  $+235\%$ . Between  $\sigma_0 = 5$  and 20 mm at 100 kW,  $\Delta \propto \sigma_0^{-1.84 \dots -2.01}$ , the impulsive  $\sigma_0^{-2}$  of Eq. (3) rather than the continuous-beam  $\sigma_0^{-1.5}$ . Aligned extraction  $H_2^+$  widths at 0 G and 100 kW are listed in Table 1. A 3 mm extraction beam is already unusable as a size measurement at 100 kW. The most severe case remains injection at 3 mm and 500 kW, where  $\sigma_m \approx 56$  mm ( $+1700\%$ ) for all three species.

**Table 1:** Aligned extraction,  $H_2^+$ , 0 G, 100 kW, 25 kV: obtained width and expansion versus true size.

| $\sigma_0$ (mm) | 3     | 5    | 7    | 10   | 15   | 20   |
|-----------------|-------|------|------|------|------|------|
| $\sigma_m$ (mm) | 38.2  | 26.8 | 22.4 | 20.4 | 21.7 | 24.9 |
| $\Delta$ (%)    | +1172 | +435 | +219 | +104 | +44  | +24  |

Species ordering follows the mixed regime. At injection, 10 mm, 0 G and 100 kW the expansions are  $+92\%$  ( $H_2^+$ ),  $+65\%$  ( $H_2O^+$ ) and  $+56\%$  ( $N_2^+$ ): lighter ions expand more, but less than a pure  $M^{-1/2}$



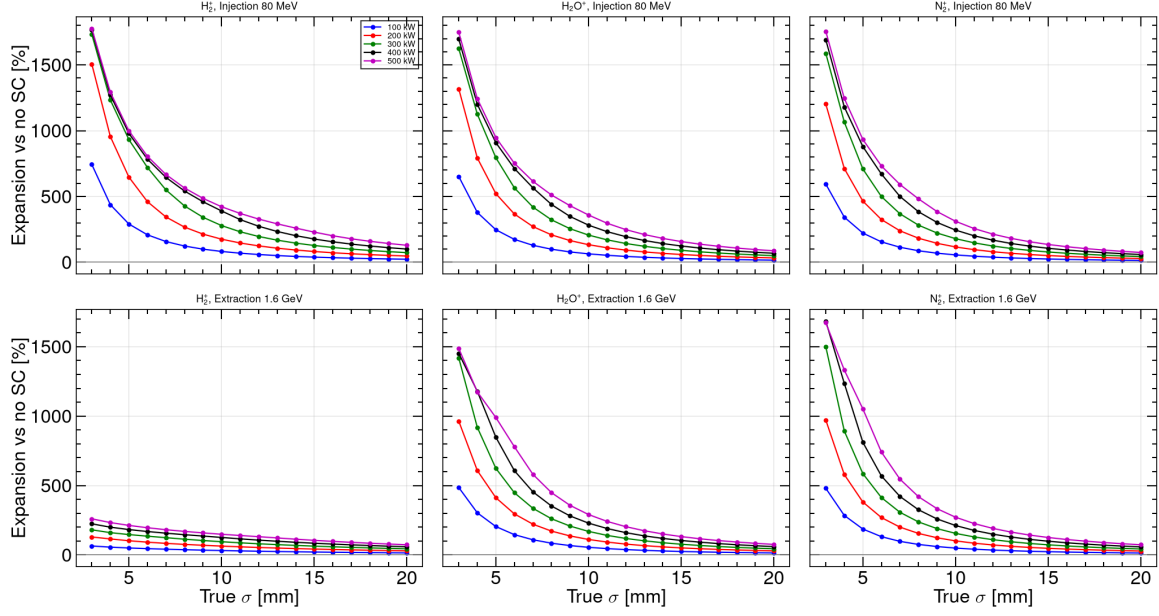
**Fig. 2:** Ion-mode obtained width  $\sigma_m$  versus true width  $\sigma_0$  at 0.1 T and 100 kW. The dashed line is  $\sigma_m = \sigma_0$ . Space charge inflates every point.

law because  $\text{H}_2^+$  leaves during the 120 ns bunch. Painted injection (25 mm) is milder (+17%, +10%, +9%). A second extraction bunch, still in the cage when the heavy ions arrive, adds a few points of extra width (single-bunch +36% and +29% versus three-bunch +41% and +40% for  $\text{H}_2\text{O}^+$  and  $\text{N}_2^+$ );  $\text{H}_2^+$  does not see that bunch (+104% either way). At commissioning power, aligned extraction  $\text{H}_2^+$  (10 mm, 0 G) is already +19% at 20 kW and +82% at 80 kW. These magnitudes are of the same order as the operational ion-mode biases reported at J-PARC RCS (20–30% wider than the MWPM) [11] and at the Fermilab Booster [8].

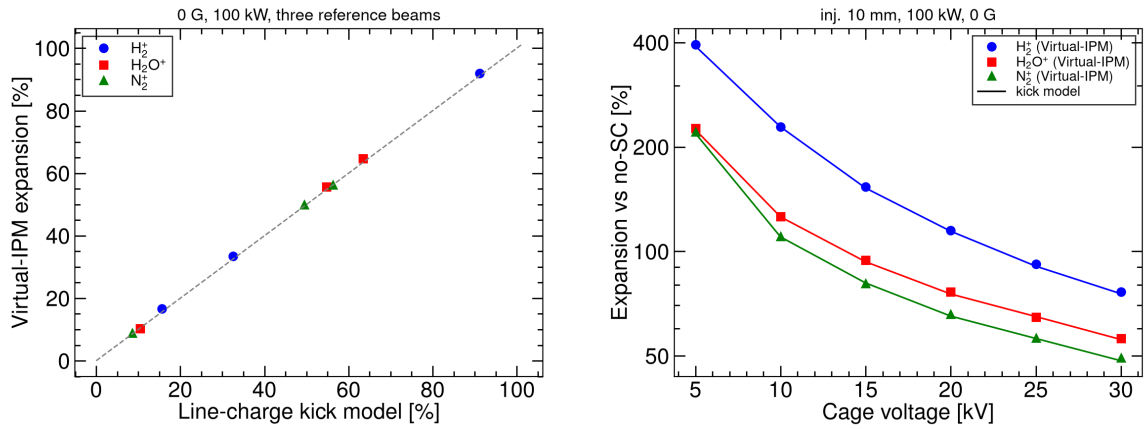
## 4 Mitigation

### 4.1 Magnetic confinement of electrons

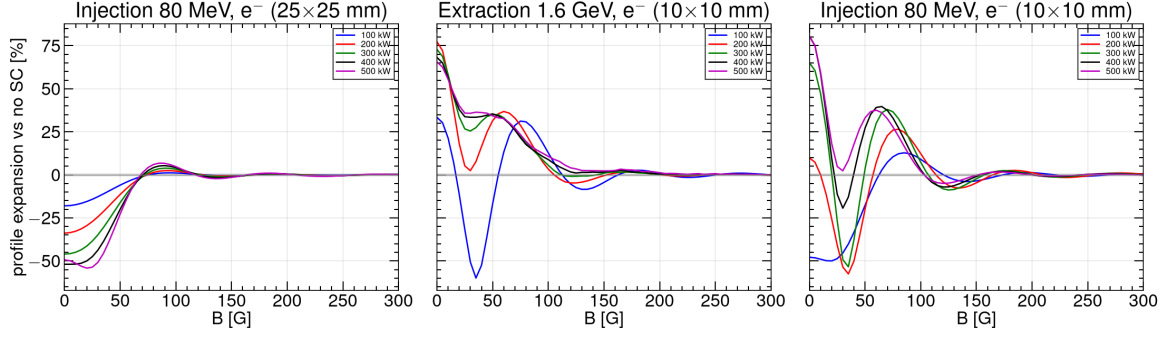
Figure 5 shows the electron expansion versus  $B$  for the three reference beams and 100–500 kW. The curves oscillate through focusing and over-kick. Painted injection recovers first; extraction and the 10 mm injection core still oscillate through 150–250 G. At 300 G every family has settled inside  $\pm 1\%$  (Fig. 6).



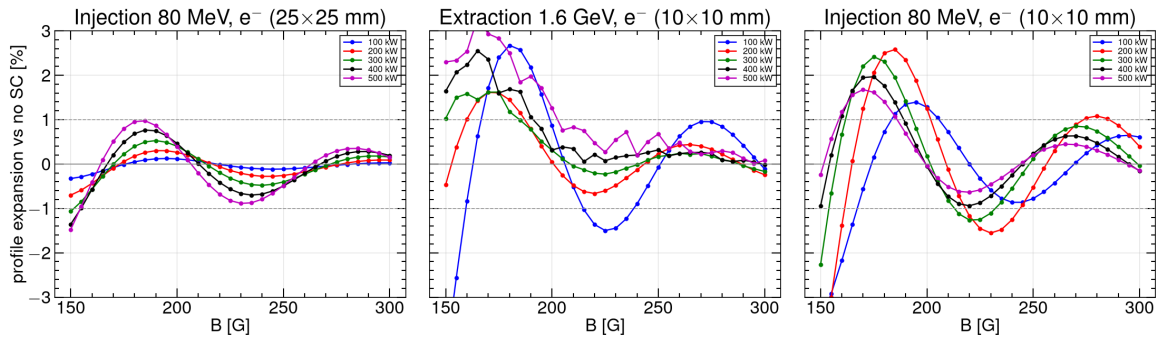
**Fig. 3:** Ion-mode expansion versus true size at 0.1 T for beam powers from 100 to 500 kW. The 3 mm/500 kW family saturates on the cage wall ( $\sigma_m \approx 56$  mm).



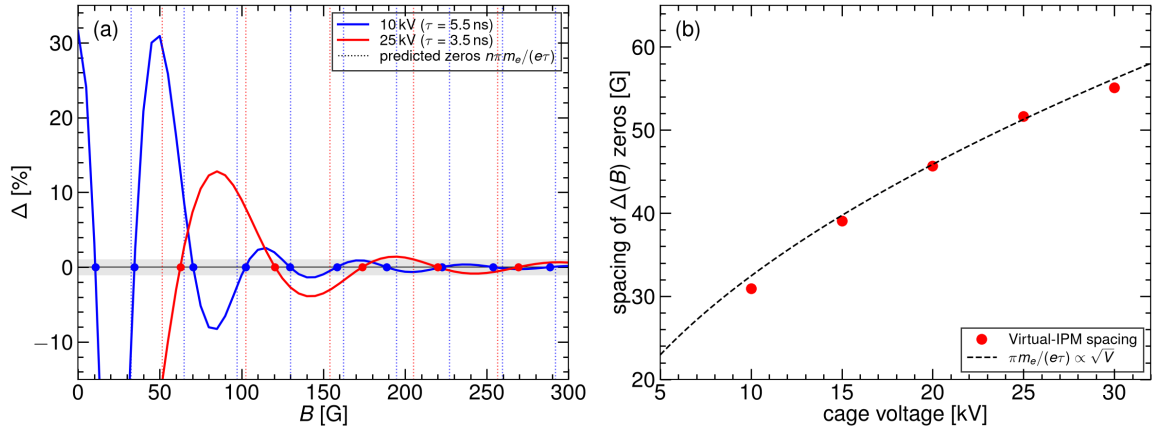
**Fig. 4:** Left: ion-mode expansion at 0 G and 100 kW, reduced kick model versus Virtual-IPM. Right: cage-voltage scan at 0 G. Agreement is  $\pm 1\%$  absolute.



**Fig. 5:** Electron-mode expansion versus guiding field, 0–300 G, for painted injection (25 mm), extraction (10 mm) and the 10 mm injection core, at 100–500 kW. The grey band is  $\pm 1\%$ .



**Fig. 6:** As Fig. 5, 150–300 G,  $\pm 3\%$  vertical scale. At 300 G every family lies inside  $\pm 1\%$ .



**Fig. 7:** (a) Relative expansion  $\Delta$  versus guiding field at two cage voltages (100 kW, 10 mm injection). Dotted vertical lines (colour-matched to the curves) are the predicted zeros  $B_n = n\pi m_e/(e\tau)$ ; circles are the simulated crossings. The grey band is  $\pm 1\%$ . The 25 kV point at  $B = 0$  ( $\Delta = -48\%$ ) is off-scale. (b) Mean spacing of those zeros versus cage voltage, compared with  $\pi m_e/(e\tau) \propto \sqrt{V}$  (no free parameter).

Figures 5 and 6 do not fall monotonically:  $\Delta(B)$  oscillates through focusing ( $\Delta < 0$ ) and over-kick ( $\Delta > 0$ ). Figure 7 identifies that oscillation as the cyclotron phase of a nearly impulsive space-charge kick.

An electron born on a 10 mm beam leaves the bunch in  $\lesssim 1$  ns and then drifts the remaining  $\tau \simeq 3.5$  ns (25 kV) in the uniform cage field. The transverse impulse  $\Delta v$  acquired inside the bunch is converted into cyclotron motion at  $\omega_c = eB/m_e$ . The displacement at the collector is  $(\Delta v/\omega_c) \sin(\omega_c \tau)$  and therefore vanishes whenever the electron completes an integer number of half-turns,  $\omega_c \tau = n\pi$  ( $n = 1, 2, \dots$ ). Those fields are exactly the equally spaced zeros  $B_n = n\pi m_e/(e\tau)$  of Eq. (2). Because  $\tau \propto V^{-1/2}$  in a uniform field, the spacing itself scales as  $\sqrt{V}$ . The amplitude of each successive lobe falls as  $1/B$  (the  $1/\omega_c$  prefactor). Magnetic mitigation is the decay of that envelope, not the choice of a particular zero: a zero can be moved by a few kilovolts or a change of orbit, whereas the envelope is a property of  $B$ .

Panel (a) is a 5 G-step scan at 100 kW and 10 mm injection, drawn at 10 and 25 kV. Vertical dotted lines are the predicted zeros  $B_n = n\pi m_e/(e\tau)$  for each voltage; filled circles mark the Virtual-IPM crossings. The 10 kV curve oscillates faster because the flight is longer ( $\tau = 5.51$  ns versus 3.48 ns), so a given phase  $n\pi$  is reached at a weaker  $B$ . The first 25 kV point,  $\Delta = -48\%$  at  $B = 0$ , lies below the frame. At 25 kV the remaining lobes sit near 85, 140, 195, 240 and 295 G with amplitudes 13, 4, 1.4, 0.9 and 0.6%.

Panel (b) extracts the mean spacing between successive zeros at five cage voltages and compares it with  $\pi m_e/(e\tau)$  (Table 2). The points follow the  $\sqrt{V}$  curve to  $\leq 5\%$  from 10 to 30 kV with no free parameter. That is the evidence that the oscillation in Figs. 5 and 6 is the cyclotron phase, and it is why 300 G—the first field at which the envelope itself lies inside  $\pm 1\%$  at every power—is the operating point rather than any particular zero.

Table 3 lists the smallest field after which  $|\Delta|$  remains below 1%, together with the residual at 300 G. Painted injection is restored first. The last curve to settle is the 10 mm injection core at 200 kW (290 G). A field of 300 G keeps every power at  $|\Delta| \lesssim 1\%$ .

The same 300 G point holds when the remaining axes are varied. From 200 G the obtained-versus-true plot lies on the diagonal except at 3 mm (Fig. 8). At 300 G, injection at 3 mm is still  $+1\%$  (100 kW) to  $+4\%$  (500 kW); 0.1 T places 3, 10 and 20 mm on the diagonal at every power ( $|\Delta| \leq 0.05\%$ ). At



**Table 2:** Electron time of flight from the cage mid-plane and the spacing of  $\Delta(B)$  zeros (5 G voltage scan, injection 10 mm, 100 kW). The prediction is Eq. (2) with  $\tau = \sqrt{2dm_e/(eE)}$  and  $d = 115.5$  mm.

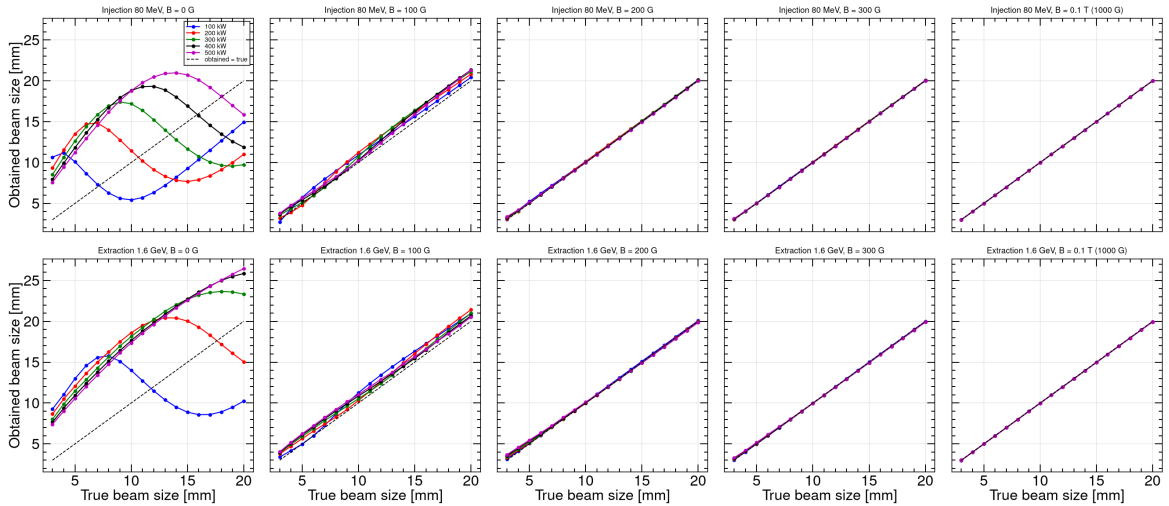
| Cage voltage | ToF $\tau$ | $\Delta B$ predicted | $\Delta B$ simulated |
|--------------|------------|----------------------|----------------------|
| 10 kV        | 5.51 ns    | 32.4 G               | 30.9 G               |
| 15 kV        | 4.50 ns    | 39.7 G               | 39.1 G               |
| 20 kV        | 3.89 ns    | 45.9 G               | 45.7 G               |
| 25 kV        | 3.48 ns    | 51.3 G               | 51.7 G               |
| 30 kV        | 3.18 ns    | 56.2 G               | 55.1 G               |

**Table 3:** Smallest guiding field after which  $|\Delta| \leq 1\%$  for the remainder of the 0–300 G grid. The residual at 300 G is given in parentheses.

| Beam       | 100 kW         | 200 kW         | 300 kW         | 400 kW         | 500 kW         |
|------------|----------------|----------------|----------------|----------------|----------------|
| Inj. 25 mm | 105 G (+0.03%) | 115 G (+0.09%) | 155 G (+0.16%) | 155 G (+0.19%) | 155 G (+0.15%) |
| Ext. 10 mm | 240 G (−0.11%) | 190 G (−0.24%) | 185 G (−0.17%) | 195 G (−0.02%) | 205 G (+0.08%) |
| Inj. 10 mm | 210 G (+0.61%) | 290 G (+0.40%) | 235 G (−0.04%) | 190 G (−0.16%) | 190 G (−0.15%) |

300 G and 10–25 kV, both 100 and 500 kW remain  $\lesssim 1\%$ . Higher  $V$  requires slightly more  $B$  at 100 kW (150 G at 10 kV to 210 G at 25 kV) because  $\tau$  is shorter and the first lobes move out; 500 kW recovers earlier at every voltage. Along the energy ramp, 300 G and 100 kW give  $|\Delta| \leq 0.5\%$  from 80 MeV to 1.6 GeV for both 120 ns and 20 ns bunches. At commissioning powers (20/50/80 kW) and 300 G the residuals are +0.00/+0.01/+0.02% (painted injection), +0.28/+0.56/+0.18% (extraction 10 mm) and +0.11/+0.29/+0.49% (injection 10 mm).

A few millimetres of vertical orbit offset change  $\Delta$  by a few percent at  $B = 0$  and by  $\lesssim 1\%$  at 300 G; the 1% threshold moves by at most  $\sim 10$  G. A horizontal offset is a translation of a uniform cage and does not change the width. The space-charge-induced centroid shift falls as  $1/B$  (the  $\mathbf{E} \times \mathbf{B}$  / polarisation drift of Ref. [7]): 0.9 mm at 100 G to 0.2 mm at 300 G on the extraction beam.



**Fig. 8:** Electron-mode obtained width versus true width. From 200 G the points lie on the diagonal; 0.1 T includes the 3 mm beams.

Thus  $B = 300$  G reduces an electron-mode error of tens of percent (and the wrong sign at 100 kW and high  $V$ ) to  $\lesssim 1\%$ . The design 0.1 T provides a factor-of-three margin and is the appropriate field for a 3 mm beam. The CSNS magnet (0.2 T) is not the limiting device. The SNS ring IPM was designed



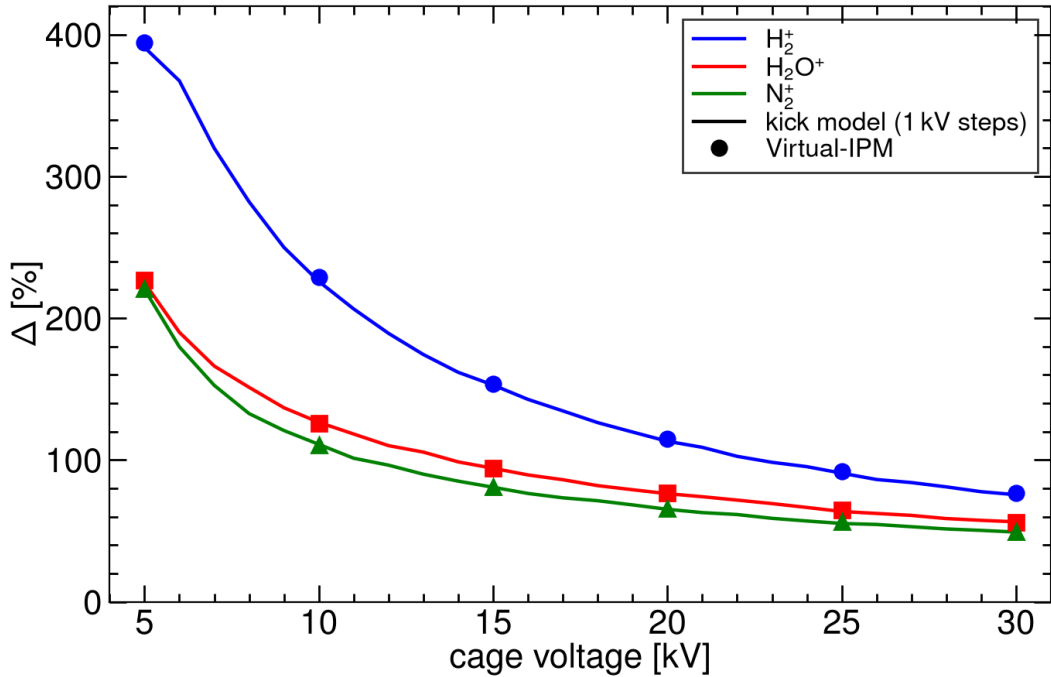
for 300 G at twenty-five times this bunch charge, with an estimated residual of about 7% [12]; a residual  $\lesssim 1\%$  at CSNS is the expected scaling. The CERN PS beam-gas ionization monitor operates at 200 mT for LHC-type bunches [13], at the high-field end of the same scaling.

## 4.2 Cage voltage

For electrons at  $B = 0$ , voltage is not a size correction: it can reverse the sign of  $\Delta$  at 100 kW and cannot reverse it at 500 kW (Section 3.1). Once  $B \geq 300$  G the design 25 kV already lies inside 1%, so the voltage may be chosen for collection efficiency and detector operation.

For ions, a higher  $V$  shortens  $\tau_2$  and reduces the integrated kick, as at ISIS (broadening  $\propto 1/E$ ) [9]. The relevant ion-mode scan is therefore  $\Delta(V)$ , not  $\Delta(B)$ : even at the design 0.1 T the ion cyclotron phase is  $\omega_c \tau_2 \lesssim 1$  rad, so a guiding field is not a size correction.

Figure 9 is that voltage scan for the three time-of-flight-resolved species on the 10 mm injection beam at 100 kW and  $B = 0$ . Curves are the line-charge kick model at 1 kV steps from 5 to 30 kV; markers are the Virtual-IPM Block C points on the 5 kV grid. The model tracks every marker to  $\leq 4\%$  relative ( $\leq 4$  points absolute). Expansion remains positive at every voltage: from 5 to 30 kV,  $\text{H}_2^+$  falls from +395% to +77%,  $\text{H}_2\text{O}^+$  from +227% to +56%, and  $\text{N}_2^+$  from +221% to +49% (Table 4). The Virtual-IPM points scale as  $V^{-0.92}$  ( $\text{H}_2^+$ ),  $V^{-0.77}$  ( $\text{H}_2\text{O}^+$ ) and  $V^{-0.83}$  ( $\text{N}_2^+$ ). Aligned extraction  $\text{H}_2^+$  (10 mm, 100 kW) falls from +280% at 5 kV to +104% at 25 kV and +93% at 30 kV. Extrapolating  $V^{-0.92}$  from the 25 kV injection point, a 10% bias would require approximately 280 kV across the cage, which is not a practical path. ESS can operate an ion IPM at 300 kV  $\text{m}^{-1}$  because its bunches are weaker by about 5000 [14]; that choice does not transfer to CSNS.



**Fig. 9:** Ion-mode expansion versus cage voltage for  $\text{H}_2^+$ ,  $\text{H}_2\text{O}^+$  and  $\text{N}_2^+$  (injection 10 mm, 100 kW,  $B = 0$ ). Curves: kick model at 1 kV steps. Markers: Virtual-IPM. Guiding  $B$  is not scanned.

Voltage is therefore a useful lever on the ion bias (a factor of about five for  $\text{H}_2^+$  from 5 to 30 kV) but not a substitute for correction or for electron-mode operation.

**Table 4:** Ion-mode expansion versus cage voltage at  $B = 0$  (injection 10 mm, 100 kW). Virtual-IPM Block C.

| $V$ (kV) | 5    | 10   | 15   | 20   | 25  | 30  |
|----------|------|------|------|------|-----|-----|
| $H_2^+$  | +395 | +229 | +153 | +115 | +92 | +77 |
| $H_2O^+$ | +227 | +126 | +94  | +77  | +65 | +56 |
| $N_2^+$  | +221 | +110 | +81  | +66  | +56 | +49 |

### 4.3 Closed-orbit offset

Horizontal offsets of +5 or +10 mm leave both electron and ion widths unchanged at 100 kW: the kick is centred on the beam, in agreement with the ISIS statement that a linear scaling correction holds within about 10 mm of the axis [9]. A vertical offset of  $\pm 5$  mm changes the drift length and therefore  $\tau$ ; the ion expansion moves by a few percent, in quantitative agreement with  $\sigma_m^2 - \sigma_0^2 \propto d$  (0.5% absolute). Electron recovery at 300 G is preserved ( $|\Delta| \leq 0.65\%$  out to  $\pm 10$  mm). At 500 kW an uncorrected ion profile ( $\sigma_m \approx 52$  mm) is clipped by the cage, and a 10 mm horizontal offset then pulls the centroid by 4 mm. That is an aperture effect, not a failure of the electron-mode operating point.

### 4.4 Inversion of ion-mode widths

Because the ion bias is large, smooth, and reproduced by a one-dimensional kick model (Fig. 4),  $\sigma_0$  can be recovered from  $\sigma_m$  without changing the hardware. The procedure is that used on the Fermilab Booster [8]: identify the time-of-flight peak; read  $\sigma_m$  and the DCCT bunch charge  $N$ ; invert  $\sigma_m = \sigma_0 (1 + \Delta)$  using a table  $\Delta(\sigma_0, N, V, \text{species})$  generated by the kick model or by Virtual-IPM, starting from  $\sigma_0 = \sigma_m$ . CSNS has the same inputs plus resolved time-of-flight peaks. The extraction column must correspond to ions born in time with the bunch (Fig. 10;  $H_2^+$  +104% at 10 mm, 100 kW, 0 G). Reported accuracies on the Booster are 5–10% on  $\sigma_0$ .

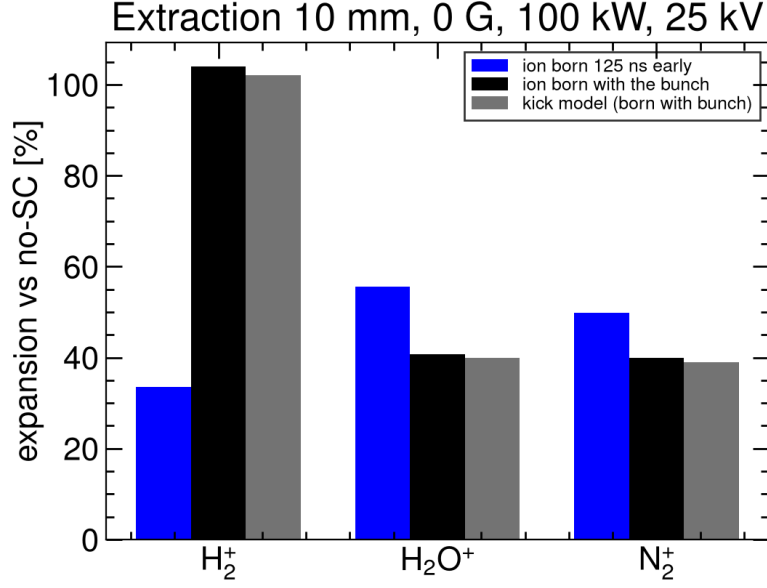
The inversion assumes that the profile has not reached the cage. At 3 mm and 500 kW it has, and no table recovers  $\sigma_0$  from a wall-saturated r.m.s. width. Field-cage non-uniformity—J-PARC reported a factor-of-two shrink from the external map alone [11]—is not included and will have to be folded in once the CSNS measured map is available.

### 4.5 Summary

Table 5 collects the efficiencies. Magnetic confinement at 300 G is the only mitigation that returns a faithful electron size. Cage voltage and look-up inversion address ion-mode bias; neither brings an uncorrected 10 mm, 100 kW ion profile to the 10% level.

**Table 5:** Efficiency of the mitigations examined. Residuals are quoted for the families of Sections 3 and 4.

| Mitigation                                        | Electron mode                                                                           | Ion mode                                                                                            |
|---------------------------------------------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| $B = 300$ G                                       | tens of percent $\rightarrow \lesssim 1\%$ (all $P$ , $V$ , energy; $\sigma \geq 4$ mm) | —                                                                                                   |
| Design $B = 0.1$ T                                | 3 mm on the diagonal; residual $\leq 0.05\%$                                            | —                                                                                                   |
| Raise $V$ (5 $\rightarrow$ 30 kV)                 | not a size fix at $B = 0$ ; unused once $B \geq 300$ G                                  | +395% $\rightarrow$ +77% ( $H_2^+$ , inj. 10 mm, 100 kW, $B = 0$ ); 10% would require $\sim 280$ kV |
| Orbit $\lesssim 10$ mm                            | 300 G still $\lesssim 1\%$                                                              | $\Delta x$ : none; $\Delta y$ : a few percent                                                       |
| Invert $\sigma_m$<br>$\sigma_0 h(\sigma_0, N, V)$ | = unnecessary at $\geq 300$ G                                                           | recovers $\sigma_0$ if $N$ and species are known and the profile is inside the cage                 |



**Fig. 10:** Extraction, 10 mm, 0 G, 100 kW, 25 kV. Virtual-IPM with the ion born in time with the bunch matches the kick model. These expansions are the extraction ion widths to be inverted.

## 5 Conclusions

Space-charge distortion of the CSNS RCS IPM has been quantified with uniform-field tracking and checked against a reduced kick model and against the published electron-mode and ion-mode scalings.

Electrons experience an attractive kick that can focus or over-focus. The resulting  $\Delta(B)$  is a cyclotron-phase oscillation whose envelope falls below 1% at 300 G for every beam, power, voltage and offset examined, and along the energy ramp. Size growth with charge, and the non-monotonic  $\sigma_m(\sigma_0)$  at  $B = 0$ , both disappear at that field. The design 0.1 T is a margin, except for the smallest (3 mm) beams.

Ions experience a repulsive kick and are always collected with  $\sigma_m > \sigma_0$ . The inflation scales as  $N/\sigma_0^2$  (impulsive) and as approximately  $1/V$ , and matches the kick model to 1%. Raising the cage voltage cannot bring a 10 mm, 100 kW ion profile to the 10% level. A look-up inversion can, provided the profile has not reached the wall.

The recommended operating point is therefore electron collection at  $B \gtrsim 300$  G. Ion time-of-flight should be used for species identification; ion-mode widths should be reported only after inversion. Once space charge is removed, the remaining electron-mode systematic is hardware: microchannel-plate saturation after  $\sim 200 \mu s$ , electromagnetic interference, secondary electrons, and the real cage map [2, 3, 5].

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The hardware parameters follow Ref. [4]. Tracking was performed with Virtual-IPM 2.3.1 [6]. This note is prepared for internal review.

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